Chain Complex Structure on the 4D Hypercubic Lattice: Categorical Isomorphism between Kihara Cube-Packing and Wilson Lattice Gauge Theory

(Paper 8: α-Identity II)

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Abstract

This paper clarifies the mathematical positioning of the self-consistent identity $\alpha^{-1} = N(1) + V_4(1) \cdot \alpha = 137 + (\pi^2/2)\alpha$ (predicting α to 8.7 ppb precision) observed in the companion paper (Paper 7: α-Identity). Paper 7 started from a purely geometric approach (4D cube-packing) but reached a precision too high to be coincidence. Reviewing the relationship to standard gauge theory, we have found that **the geometry of the 4D hypercubic lattice and Wilson lattice gauge theory (Wilson 1974) are categorically isomorphic at the level of chain complex, symmetry group, and discrete Stokes structure**.

In this paper, we (i) define the standard chain complex $C_\bullet = (C_0, C_1, C_2, C_3, C_4)$ on the 4D Euclidean integer lattice $\mathbb{Z}^4$, (ii) show that Schläfli duality (tesseract $\leftrightarrow$ 16-cell) provides the isomorphism $C_n \leftrightarrow C_{4-n}$, (iii) demonstrate that the hypercubic symmetry group B_4 acts equivariantly on both, and (iv) show that the discrete Stokes theorem produces identical bulk-boundary decompositions. Wilson theory is the formalism that places gauge fields on C_1 (links) and an action on C_2 (plaquettes); Kihara cube-packing is the formalism that places volumes on C_4 (4-cells) and measures on C_3 (boundary faces). They are **categorically isomorphic** via Schläfli duality.

This isomorphism opens (a) gauge-theoretic computational tools (Wilson strong-coupling expansion, renormalisation group, Monte Carlo, etc.) for application to the Kihara programme, and (b) conversely, Kihara-side number-theoretic identities (Lagrange–Jacobi four-square theorem, inclusion-exclusion) as new computational methods for Wilson theory. Paper 7's inside/outside decomposition $1 = 137\alpha + V_4(1)\alpha^2$ is naturally interpreted as a bulk-boundary normalisation condition on this chain complex.

A first-principles derivation of the numerical value of α, the privilege of $R = 3$, and the geometric identification of the residual $c_3 \approx 1.6 \times 10^{-3}$ are left as open problems beyond this paper's scope. The contribution of this paper is to demonstrate, in standard mathematical language, that Paper 7 is not mere numerology but **a natural manifestation of the topological structure of the 4D lattice**.

1 Introduction

1.1 Reach of Paper 7 and Motivation

In the companion paper (Paper 7, Concept DOI: 10.5281/zenodo.19869266, v3), we observed the self-consistent algebraic identity

$$\alpha^{-1} = N(1) + V_4(1) \cdot \alpha = 137 + \frac{\pi^2}{2}\alpha \tag{1.1}$$

constructed from $N(1) = 137$ (the count of integer-centred unit cubes fully contained in a 4D ball of radius 3) and $V_4(1) = \pi^2/2$ (the volume of the 4D unit ball). Solving the quadratic $(\pi^2/2)\alpha^2 + 137\alpha - 1 = 0$ for α yields $\alpha = 7.29735194 \times 10^{-3}$, agreeing with the CODATA 2018 value to a relative error of 8.7×10^{-8} (8.7 ppb).

This precision is approximately 1/3000 of the Eddington-style integer-fitting deviation (2.6×10^{-4}), too good to be dismissed as "mere numerical coincidence". Paper 7 was carefully crafted to evade the Eddington trap by emphasising (i) the self-consistent equation structure, (ii) the independent 4D-geometric derivation of both 137 and $\pi^2/2$, and (iii) the QFT-perturbative analogy. However, the physical mechanism and the privilege of $R = 3$ remained open.

The motivation of this paper is the suspicion, given the strikingly high precision, that there must be a deeper mathematical structure underlying the identity. Specifically, on reviewing standard lattice gauge theory established by Wilson (1974), we recognise that the geometric setup of Paper 7 and Wilson's lattice formulation are **almost identically isomorphic from a geometric perspective**. The aim of this paper is to make this isomorphism precise in categorical language.

1.2 Scope of the Claim

This paper **claims** the following:

1. The standard chain complex on the 4D Euclidean integer lattice $\mathbb{Z}^4$ admits an equivariant action of the hypercubic symmetry group B_4.
2. Schläfli duality (the self-duality of the tesseract / 16-cell) provides the isomorphism $C_n \leftrightarrow C_{4-n}$.
3. Wilson lattice gauge theory and Kihara cube-packing are formalisms that place different structures on this chain complex, and they are **categorically isomorphic** via Schläfli duality.
4. The discrete Stokes theorem produces an identical bulk-boundary decomposition in both formalisms.
5. Paper 7's inside/outside decomposition $1 = 137\alpha + V_4(1)\alpha^2$ is naturally interpreted from this isomorphism structure.

This paper does **not** claim:

1. That the value of α is derived from first principles via this isomorphism (we cite Paper 7 as observation only)
2. That the privilege of $R = 3$ is proved by this isomorphism

3. That U(1) gauge symmetry is uniquely emergent from this framework
4. The geometric identification of $c_3 \approx 1.6 \times 10^{-3}$
5. The specific connection to Higgs mechanism or electroweak symmetry breaking

These are all explicitly stated as open problems in §8.

2 Chain Complex on the 4D Hypercubic Lattice

2.1 Lattice Definition

Denote the 4D Euclidean integer lattice by $\Lambda = \mathbb{Z}^4 \subset \mathbb{R}^4$. Each $x \in \Lambda$ is a **site** (vertex). Directed edges between adjacent sites are **links**, with the set of links denoted L. The smallest 2D faces enclosed by links are **plaquettes**, 3D cells are **3-cells**, and 4D cells are **4-cells** (tesseracts).

2.2 Chain Complex

Free abelian groups generated by these cells with formal integer coefficients:

$$C_0 = \mathbb{Z}\langle\text{vertices}\rangle,\ C_1 = \mathbb{Z}\langle\text{links}\rangle,\ C_2 = \mathbb{Z}\langle\text{plaquettes}\rangle,\ C_3 = \mathbb{Z}\langle\text{3-cells}\rangle,\ C_4 = \mathbb{Z}\langle\text{4-cells}\rangle.$$

The boundary operators $\partial_n : C_n \to C_{n-1}$ are defined standardly (with orientation):

$$\partial_n \circ \partial_{n+1} = 0. \tag{2.1}$$

This makes $C_\bullet$ a chain complex.

2.3 Action of B_4

The hypercubic symmetry group B_4 (Coxeter group of order $|B_4| = 2^4 \cdot 4! = 384$) acts on each C_n, equivariantly with ∂_n:

$$g \cdot \partial_n = \partial_n \cdot g, \quad \forall g \in B_4. \tag{2.2}$$

2.4 f-Vector of the Tesseract

n	Unit cell of C_n	Boundary cells per unit	Per tesseract count
0	vertex	—	16
1	link	2 vertices	32
2	plaquette	4 links	24
3	3-cell	6 plaquettes	8
4	4-cell	8 3-cells	1

3 Schläfli Duality and Poincaré Duality

3.1 Self-Duality of Tesseract and 16-cell

Tesseract ($\{4,3,3\}$) and 16-cell ($\{3,3,4\}$) are mutually dual. The k-cell of the tesseract corresponds to the $(4-k-1)$-cell of the 16-cell:

Tesseract	16-cell
16 vertices ($k = 0$)	16 cells ($k = 3$)
32 edges ($k = 1$)	32 faces ($k = 2$)
24 faces ($k = 2$)	24 edges ($k = 1$)
8 cells ($k = 3$)	8 vertices ($k = 0$)

3.2 Poincaré Duality on the Chain Complex

For the dual lattice $\Lambda^* \cong \mathbb{Z}^4$ (half-integer-shifted), there is a discrete Poincaré duality:

$$D : C_n(\Lambda) \xrightarrow{\sim} C_{4-n}(\Lambda^*), \tag{3.1}$$

which is B_4-equivariant:

$$D \circ g = g \circ D, \quad \forall g \in B_4. \tag{3.2}$$

3.3 Concrete Examples of the Dual Action

Λ side	Λ^* side
Site (C_0)	4-cell (C_4)
Link (C_1)	3-cell (C_3)
Plaquette (C_2)	Plaquette (C_2, self-dual)
3-cell (C_3)	Link (C_1)
4-cell (C_4)	Site (C_0)

The self-duality $C_2 \to C_2^*$ is the direct origin of the isomorphism between Wilson plaquette action and Kihara cube-packing.

4 Wilson Lattice Gauge Theory in Chain-Complex Language

4.1 Link Variables and Gauge Transformation

Each link $\ell = (x, y) \in L$ carries an element $U_\ell = e^{i\theta_\ell} \in U(1)$. Gauge transformation:

$$U_{xy} \to e^{i\alpha(x)} U_{xy} e^{-i\alpha(y)}. \tag{4.1}$$

4.2 Plaquette Phase and Wilson Action

Plaquette phase:

$$U_p = \prod_{\ell \in \partial p} U_\ell, \tag{4.2}$$

gauge invariant under (4.1). Wilson action:

$$S_{\text{Wilson}} = \frac{1}{g^2} \sum_{p \in C_2} \left[1 - \operatorname{Re} U_p\right]. \tag{4.3}$$

4.3 Discrete Stokes Theorem

For any closed 2D surface Σ:

$$\prod_{p\in\Sigma} U_p = \prod_{\ell\in\partial\Sigma} U_\ell. \tag{4.4}$$

This is the discrete Stokes theorem, a consequence of $\partial^2 = 0$.

5 Kihara Cube-Packing in Chain-Complex Language

5.1 Unit Cube Configuration

$$\square(c) = \prod_{i=1}^{4} \left[c_i - \frac{1}{2}, c_i + \frac{1}{2}\right] \in C_4(\Lambda^*). \tag{5.1}$$

A Kihara unit cube is a 4-cell of Λ^*, equivalent to a site of Λ via D.

5.2 Packing Number and Volume Deficit

Packing condition: $\sum_{i=1}^{4}(|c_i| + 1/2)^2 \leq R^2$. By direct enumeration, $N(1) = 137$ at $R = 3$.

Volume deficit:

$$\Delta(R) = V_4(R) - N(k), \quad \Delta(R) = \frac{16\pi}{3}R^3 - 6\pi R^2 + O(R). \tag{5.2}$$

5.3 Bulk-Boundary Structure

- **Bulk**: cubes pack exactly in the interior, contributing the strict integer N_{bulk}.
- **Boundary**: cubes are partially included near $\partial B(R)$, contributing $\Delta(R)$.

This is the same structure as the discrete Stokes theorem (§4.3): bulk cancels, only boundary remains.

6 Main Theorem: Categorical Isomorphism

6.1 Definition of $\mathcal{L}_4$

Definition 6.1: $\mathcal{L}_4$ is the category of chain complexes on the 4D Euclidean integer lattice with B_4-equivariant boundary operators, and morphisms are B_4-equivariant chain maps.

6.2 Isomorphism Theorem

Theorem 6.2 (Main Theorem): As objects of $\mathcal{L}_4$,

$$(\mathcal{C}_W, \partial_\bullet, B_4) \cong_{\mathcal{L}_4} (\mathcal{C}_K, \partial_\bullet, B_4)$$

where $\mathcal{C}_W$ is Wilson's chain complex (gauge fields on C_1, action on C_2) and $\mathcal{C}_K$ is Kihara's (indicator on C_4, measure on C_3). The isomorphism is given by Schläfli duality D (§3.2).

6.3 Sketch of Proof

(i) $\Lambda \cong \Lambda^*$ via half-integer shift, B_4-equivariantly.

(ii) Schläfli duality $D : C_n(\Lambda) \to C_{4-n}(\Lambda^*)$ is the discrete Poincaré duality, commuting with ∂.

(iii) B_4 acts equivariantly on both; D is B_4-equivariant.

(iv) The discrete Stokes theorem (§4.3 and §5.3) takes the same form in both.

The detailed algebraic proof follows standard methods (Hatcher 2002). ■

6.4 Corollaries

Corollary 6.3: Wilson lattice gauge theory's analytical methods (strong-coupling expansion, Monte Carlo, RG) become directly applicable to the Kihara cube-packing problem via Schläfli duality D.

Corollary 6.4: Conversely, Kihara-side number-theoretic identities (Lagrange–Jacobi four-square theorem $r_4(N) = 8\sigma(N)$, Paper 3 §6) may provide novel closed forms for finite-volume Wilson partition functions.

7 Mathematical Positioning of Paper 7's α Identity

7.1 Naturalness of Inside/Outside Decomposition

Paper 7's decomposition:

$$1 = \underbrace{137\alpha}_{\text{inside (bulk)}} + \underbrace{V_4(1)\alpha^2}_{\text{outside (boundary)}} \tag{7.1}$$

is naturally interpreted from Theorem 6.2 as a bulk-boundary normalisation condition on the chain complex of the 4D lattice.

7.2 Wilson Action Correspondence

The formalism placing the Wilson action on Paper 7's packing problem:

$$S_{\text{Kihara-Wilson}} = \sum_{p \subset B(3)} [1 - \operatorname{Re} U_p] \tag{7.2}$$

is mathematically well-defined, encompassing Paper 7's geometric setup as a finite-volume version of Wilson theory.

7.3 Repositioning of α Identity

> Paper 7's self-consistent identity $\alpha^{-1} = N(1) + V_4(1)\alpha$ is a structural identity naturally positioned, on the chain complex structure of the 4D Euclidean integer lattice, by the categorical isomorphism between Wilson lattice gauge theory and Kihara cube-packing.

This is decisively different from Eddington-style numerical fitting. However, the proof that α's numerical value is uniquely determined from this structure is beyond this paper's scope (§8).

8 Open Problems

1. **Mechanistic derivation of α's value**: whether $\alpha = 1/137.036$ is uniquely determined from the isomorphism
2. **Privilege of $R = 3$**: why $N(1) = 137$ corresponds to physical α
3. **Geometric identification of $c_3 \approx 1.6 \times 10^{-3}$**
4. **Generalisation to other gauge groups**: $SU(2), SU(3), SU(N)$, α_s analogues
5. **Projection to physical (3+1)D spacetime**: connection via central projection (Papers 2, 6) and Wick rotation
6. **Connection with Higgs mechanism**: implementation of the longitudinal scalar beat coupling
7. **Numerical verification**: Wilson Monte Carlo, Lagrange–Jacobi applications

9 Conclusion

This paper has clarified the mathematical positioning of the self-consistent identity $\alpha^{-1} = N(1) + V_4(1)\alpha$ (predicting α to 8.7 ppb precision) observed in Paper 7, as the chain complex structure of the 4D Euclidean integer lattice.

The main result is that **Wilson lattice gauge theory and Kihara cube-packing are categorically isomorphic on the chain complex (B_4 symmetry, Schläfli duality, discrete Stokes structure) of the 4D hypercubic lattice** (Theorem 6.2). Implications:

1. Paper 7's inside/outside decomposition is naturally interpreted as a bulk-boundary normalisation on the chain complex
2. Wilson theory's analytical methods become applicable to the Kihara programme
3. Kihara-side number-theoretic identities may provide novel computations for Wilson theory
4. The α identity is not mere numerology but a natural manifestation of the topological structure of the 4D lattice

The first-principles derivation of α's specific value, the privilege of $R = 3$, and the geometric identification of c_3 are explicitly listed as open problems in §8.

The contribution of this paper is not the mechanistic derivation of α but the demonstration that Paper 7 connects naturally to standard mathematical structures (algebraic topology + lattice gauge theory). This further reinforces Paper 7's evasion of the Eddington trap, and positions this research programme to be in dialogue with standard physics and mathematics communities.

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Appendix A: f-Vector and Schläfli Duality

Tesseract ($\{4,3,3\}$): $(f_0, f_1, f_2, f_3) = (16, 32, 24, 8)$. 16-cell ($\{3,3,4\}$): $(f_0, f_1, f_2, f_3) = (8, 24, 32, 16)$. Schläfli duality: $f_k(\text{Tesseract}) = f_{3-k}(\text{16-cell})$ (Coxeter 1973).

Appendix B: Numerical Reproduction of Paper 7's α Identity (reference)

Identical to Paper 7 §3.2 (v3, 8.7 ppb precision):

```
V_4(1) = ²/2  4.9348022
N(1) = 137
Equation: 4.9348 ² + 137  - 1 = 0
Positive root:   7.29735194 × 10³
¹  137.0360110
CODATA 2018: ¹ = 137.0359991
Relative error: 8.7 × 10 (8.7 ppb)
```

For details, see Paper 7 (Concept DOI: 10.5281/zenodo.19869266).